

Ticket System v3

Complete manual — project rundown & user guide

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1) Project rundown (technical)

1.1 Purpose

Ticket System v3 is a full-stack **service desk / ticketing** application for submitting requests, assigning work to teams and personnel, tracking **SLA** targets, escalating breached items, managing **KPI** maintenance tasks, and collecting **customer verification and feedback** after resolution.

1.2 Technology stack

Layer	Technology
Application	Next.js (App Router), React , Tailwind CSS
API	Next.js Route Handlers (<code>src/app/api/**</code>)
Database	PostgreSQL with Prisma ORM
Authentication	NextAuth.js — credentials, optional Google OAuth , optional corporate OIDC (OpenID Connect)
Email	SMTP (e.g. Brevo) for resolution and verification flows

1.3 Repository layout (high level)

- `src/app/` — Pages and API routes (App Router).
- `src/components/` — Shared UI (navigation, portal shell, dashboards).
- `src/lib/` — Business logic: auth, SLA, email, ticket actions, portal accounts.
- `prisma/` — Schema, migrations, seed.
- `ecosystem.config.cjs` — **PM2** process definition for production on Node hosts.

1.4 Core concepts

- **Ticket** — Single customer issue with status (Open → In progress → Resolved → Closed, plus Pending info / Escalated as needed).
 - **Priority** — Includes a “**Set Priority Level**” (**UNSET**) default until staff sets **Low / Medium / High / Urgent**; work typically cannot move to **In progress** until priority is set (policy enforced in API/board).
 - **SLA** — First-response and resolution due times derive from **SlaPolicy** per priority; automated sweep can escalate breached unresolved tickets (Admin-triggered API).
 - **Teams & agents** — Tickets route to teams; **Personnel** work assigned queues from `/agent`.
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2) Architecture & data

2.1 Request flow (simplified)

1. User authenticates → session carries **role** and **email**.
2. Customer submits ticket → **POST /api/tickets** creates row, SLA timestamps, activity log.
3. Staff manage lifecycle → **PATCH /api/tickets/[id]** (status, priority, escalation, resolution).
4. On **Resolved**, system may email the **requestor inbox** (SMTP); customer **verifies** and submits **rating** where configured.

2.2 Notification email routing

- **Google sign-in**: notification and verification emails target the **Google account email** (`authProvider: google`).
- **Portal / credentials**: `requestorEmail` on the ticket aligns with the **portal account work email** stored in the database (with safeguards); synthetic portal identities may require an explicit notification email on intake.

2.3 Pending resolution confirmation (customers)

If a customer has a **resolved** ticket that still needs **verification and mandatory rating** (no feedback record yet), **new ticket creation may be blocked** until that flow is completed. The UI surfaces a **notification** and links to the **verification** path (e.g. `/tickets/[id]/verification`).

3) Authentication & roles

3.1 Sign-in

- **URL:** /signin
- Methods: **local credentials** (username or email + password), **Google** (if configured), **corporate SSO** (if OIDC env vars are set).

3.2 Role resolution (typical)

1. Claims from IdP profile (role, roles, groups) when present.
2. Env lists **ADMIN_EMAILS** / **AGENT_EMAILS** for bootstrap mapping.
3. Default **Customer** where no other rule applies.

3.3 Role summary

Role	Typical access
Customer	Home, Create ticket , My tickets , ticket detail, verification/rating, customer profile.
Personnel	Own assignment queue (/agent), ticket workspace for assigned items, insights as configured.
Agent	Broader operational queue/board patterns (org-dependent).
Admin / SuperAdmin	Dashboard, personnel, escalation triggers, assignment oversight, SLA sweep, reports as configured.

Head / privileged portal flags may grant coordination powers (e.g. escalation) per implementation.

4) Deployment & operations

4.1 Environment

Copy `.env.example` to `.env`. Key groups: **DATABASE_URL**, **NextAuth** secret and URL, **OIDC / Google** client settings, **SMTP (Brevo)** for mail, **APP_BASE_URL / NEXTAUTH_URL** for correct email links, optional admin email lists.

4.2 Database

- **Prisma:** `npx prisma migrate deploy` or `npx prisma db push` (depending on how the environment was baselined).
- **Seed (optional):** `npm run db:seed` for demo data.
- One-off: `npx tsx scripts/ensure-unset-priority-data.ts` if using **UNSET** priority and SLA rows need ensuring after `db push`.

4.3 Production (Node + PM2)

- Build: `npm run build`
- Start: `pm2 start ecosystem.config.cjs` (or `npm run pm2:start`)

- Reload after deploy: `npm run deploy:pm2` or `pm2 restart ticket_system_v3 --update-env`
 - See repository `DEPLOY_NGINX.md`, `DEPLOY_NGINX_WINDOWS.md`, or `DEPLOY_CPANEL.md` for host-specific steps.
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5) Customer user guide

5.1 First steps

1. Open `/signin` (or `/signup` if self-registration is enabled).
2. Use the account your organization provided, or register per policy.

5.2 Submit a ticket

1. Go to `/tickets/new`.
2. Complete **department / business unit, name**, and **issue** description.
3. If you use **Google**, emails for resolution/verification go to your **Google address**. If you use a **portal** account, notification email follows your **registered work email** (optional override only when it matches your portal email).
4. Submit; track from `/my-tickets`.

5.3 Track and participate

- Open a ticket from the list or a shared link.
- Read the **activity timeline** and **conversation** thread.
- If **new requests are blocked** because a **resolved** ticket awaits your **confirmation and rating**, open the **notification** or the linked **verification** page and complete the steps.

5.4 Resolution, verification, and rating

1. When staff marks the ticket **Resolved**, you may receive email with verify / reject links.
2. In the portal, complete **verification** and **star rating / feedback** when prompted.
3. Until mandatory steps are done, **creating another ticket** may be disabled **policy**.

5.5 Ticket statuses (customer view)

Status	Meaning (plain language)
Open	Received; queued / triaged.
In progress	Actively worked.
Pending info	Waiting on information (implementation may vary).
Escalated	Elevated for leadership or specialized handling.
Resolved	Solution delivered; you should verify and rate.
Closed	Closed after confirmation / feedback per policy.

5.6 My account

From `/customer/profile` or account entry points, update profile where allowed. **Security** changes (username, email, password) may sign you out and require a fresh sign-in.

6) Staff & personnel guide

6.1 Daily routine

1. Sign in at `/signin`.
2. Open `/agent` — orchestration board, kanban, and assigned ticket access.
3. Review `/insights` for personal or team KPI visibility as granted.

6.2 Working a ticket

1. Confirm assignment and priority (**set priority** if still **UNSET** — otherwise board/API may block **In progress**).
2. Update status honestly (drag-and-drop where enabled).
3. Use **request more information** per policy (may log activity **without** forcing “pending info” status depending on configuration).
4. Resolve with clear **resolution notes**; customer receives email to **verify**.

6.3 KPI / task hygiene

- Complete checklist items only where you are the assignee (when restricted).
 - Respect recurrence boundaries (daily / weekly / monthly cycles).
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7) Admin & SuperAdmin guide

7.1 Oversight

- `/` — Operational dashboard (role-dependent).
- `/admin/personnel` — Personnel registry and assignment readiness.
- `/admin/manual-assignment` — Assignment board where configured.
- `/admin/escalation-triggers` — Priority-linked escalation behavior.
- `/reports` — Reporting views as implemented.

7.2 SLA sweep

- `POST /api/sla/sweep` (secured; Admin context) — scans for SLA breaches and escalates/logs per policy.

7.3 Accounts

- `/admin/account-management` — Portal account administration where deployed.

8) Account security (My Account)

Staff typically use `/admin/account`.

Change username

1. Security tab → new username, current password → **Update username**.
2. You are signed out; sign in with the new username.
3. Rules: length and character constraints shown in UI.

Change email / password

Similar pattern: confirm current password; after success, **sign in again** with new credentials.

Account requests

Suspension or deletion requests may be submitted for admin review.

9) Troubleshooting

Symptom	What to try
Cannot sign in	Verify credentials, caps lock; try SSO status; confirm account active.
"Forbidden" or missing menu	Your role may not include that feature — contact Admin.
Cannot create new ticket	Complete pending resolved verification/rating on existing ticket.
Email link expired / invalid	Request staff to re-send resolution flow or open ticket in portal.
Cannot move ticket to In progress	Set priority above Set Priority Level (UNSET) .
Database / deploy errors	Check <code>DATABASE_URL</code> , run Prisma migrate or push, rebuild, restart PM2.

10) Quick reference — URLs

Path	Description
<code>/signin</code>	Sign in
<code>/signup</code>	Self-registration (if enabled)
<code>/</code>	Role-aware home
<code>/tickets/new</code>	New ticket
<code>/my-tickets</code>	Customer ticket list

/tickets/[id]	Ticket detail
/tickets/[id]/verification	Resolution verification
/tickets/[id]/rate	Rating / feedback
/agent	Staff board / queue
/agent/tickets/[id]	Staff ticket workspace
/insights	KPI / insights
/process	Process / lifecycle info
/reports	Reports
/admin/personnel	Personnel admin
/admin/account	Account & security

End of document.